

### Assignment 04

Second Year BS(Honours) 2025-2026

Course Title: Math Lab II Course Code: AMTH 250

Department of Applied Mathematics, University of Dhaka

**Name:**

**Roll:**

**Group:**

*[ Write a Fortran Program to solve each of the following problems. Use files for input/output unless specified otherwise. Name the files and the code according to the assignment and problem no., e.g., for problem no. Y of Assignment X, input and output file names must be 'inXqY.txt' and 'outXqY.txt' respectively.]*

1. A hospital flags a patient as urgent 'if at least two of the three symptoms: fever, breathing trouble, and chest pain are present'. A doctor proposes an alternative rule: 'flag the patient if fever and breathing trouble together, or if chest pain is present with at least one of the other two symptoms'. Write the both rule as a logical expressions, construct the truth table, and determine whether the two rules are logically equivalent.
2. A smart thermostat decides whether to turn on the AC based on: 'the room is warm' and 'someone is home'. The manufacturer states two versions of the rule. *Version A*: Turn on the AC only if it's warm and someone is home; never run it unless both hold. *Version B*: Don't turn on the AC whenever it's not warm, or whenever nobody's home. Translate Version A and Version B into logical expressions. Then write a FORTRAN program to build the truth table for both expressions and determine whether  $(A) \leftrightarrow (B)$  is a tautology.
3. Write a FORTRAN program to construct a truth table for  $p \rightarrow q$  and  $p \leftrightarrow q$ . Also show that  $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$ .
4. Write a FORTRAN program to prove that  $\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$ . Use subroutine to calculate the corresponding biconditionals.
5. Let  $H_n$  denotes the number of moves needed to solve the Tower of Hanoi problem with  $n$  disks. The recurrence relation with the initial condition is given for the problem is  $H_n = 2H_{n-1} + 1; H_1 = 1$ . Using the recursive function as a subprogram, compute  $H_n$  for  $n = 1, 2, \dots, 10$ . If each moves takes 3 seconds and cost \$0.02, tabulate  $n, H_n$ , time(in minutes), and cost.
6. The Mersenne numbers are defined by  $M_n = 2^n - 1$  where  $n = 0, 1, 2, \dots$ . Generate the first 15 Mersenne numbers and determine which of them are prime.
7. Using the recursive function from Q5 and the formula from Q6, write a FORTRAN program that computes  $H_n$  and  $M_n$  for  $n = 1, 2, \dots, 10$ , prints them side by side, and checks whether  $H_n = M_n$  for every  $n$ . State your conclusion about the relationship between the two sequences.

8. An epidemiologist models the spread of a rumor in a village of fixed population  $N$  using the recurrence  $I_n = I_{n-1} + \frac{kI_{n-1}(N-I_{n-1})}{N}$ , where  $I_n$  is the number of people who have heard the rumor after  $n$  days. For  $N = 1000$ ,  $k = 0.3$ , and  $I_0 = 5$ , use a recursive function to compute  $I_n$  for  $n = 1, \dots, 20$  days and tabulated  $I_n$  along with the percentage of the village that has heard the rumor.
9. A warehouse robot stacks cube-shaped crates into a pyramid. The  $n$ -th layer, counted from the top contains  $n^2$  crates. Thus the total number of crates in an  $n$ -layer pyramid is  $T_n = T_{n-1} + n^2$  with  $T_1 = 1$ . Each crate costs  $p$  dollars to produce. Use a recursive function as a subprogram to compute  $T_n$  and the total cost  $C_n = C_{n-1} + pn^2$  with  $C_1 = p = \$15$ . Create a table for  $n = 1, \dots, 20$  with the following columns: layer ( $n$ ), crates in the  $n$ -th layer ( $n^2$ ), total crates ( $T_n$ ), and total cost ( $C_n$ ).
10. Consider the recurrence relation  $a_n = 2a_{n-1} - a_{n-2} + 3$  with the initial conditions  $a_0 = 1$ ,  $a_1 = 5$ . Find  $a_n$  for  $n = 1, \dots, 50$ . Compare the results with the exact solution  $a_n = \frac{3}{2}n^2 + \frac{5}{2}n + 1$ . Show your results with suitable descriptor making a suitable table.
11. We know from the **Binomial Theorem** that:

$$(a + b)^n = C(n, 0)a^n b^0 + C(n, 1)a^{n-1}b^1 + C(n, 2)a^{n-2}b^2 + \dots + C(n, n)a^0 b^n$$

From this expansion, the general  $(n + 1)^{th}$  term can be expressed as:

$$T_{n+1} = C(n, r)a^{n-r}b^r$$

Use a recursive function as a subprogram to compute factorial values, and then apply this function to determine the binomial coefficient  $C(n, r)$ . Next construct a table in the format shown below for the expansion of  $(a + b)^n$ . For example let  $a = 3$ ,  $b = 2$ ,  $n = 5$ , then the table will be

| $r$      | Term $T_{n+1}$ | Coefficient $C(n, r)$ | Value of term $C(n, r)a^{n-r}b^r$ |
|----------|----------------|-----------------------|-----------------------------------|
| 0        | 1              | 1                     | 243                               |
| 1        | 2              | 5                     | 810                               |
| 2        | 3              | 10                    | 1080                              |
| $\vdots$ | $\vdots$       | $\vdots$              | $\vdots$                          |